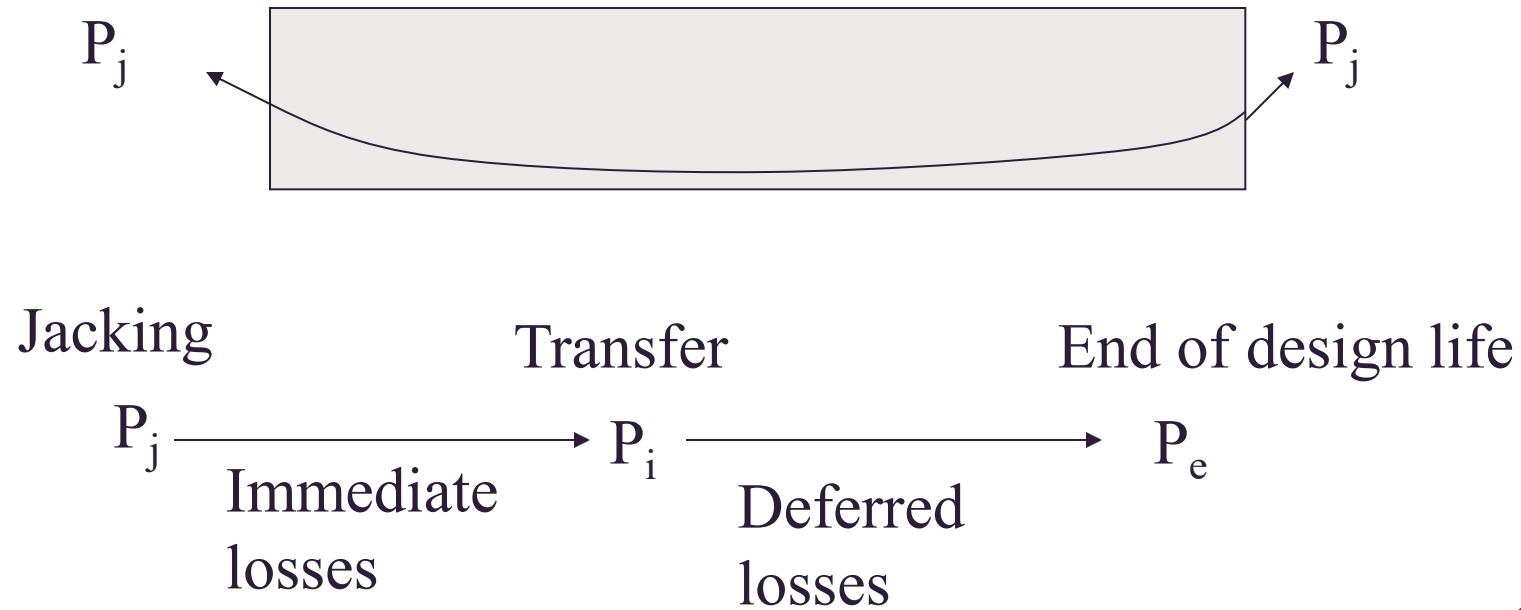
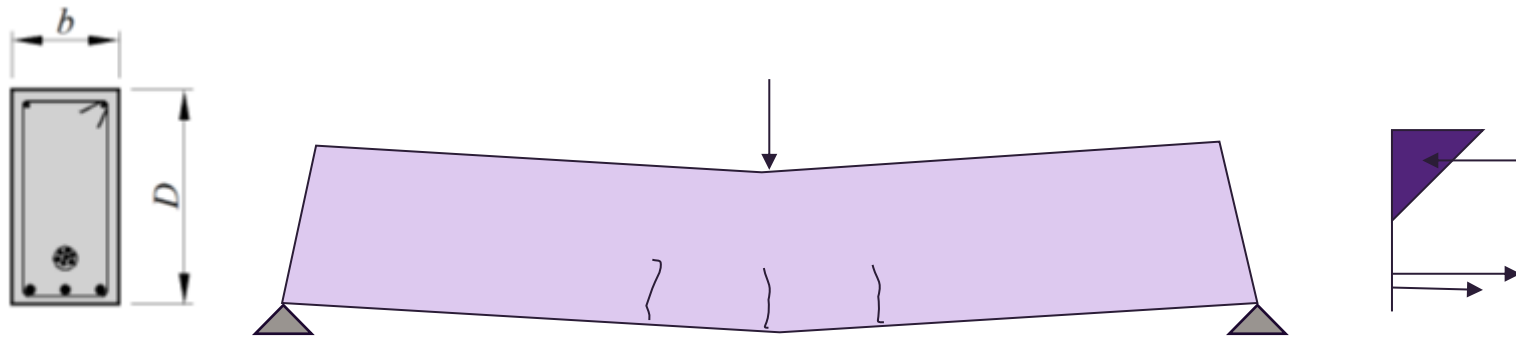


Week 4 - Losses of Prestress

Chapter 9 – Text book Warner et al.



Flexural Behaviour in Post-cracking range



$$I \neq I_g, \quad A \neq A_g$$

Cracking moment, Clause 8.5.3.1 of AS3600

Applied moment at which the bottom fibre of the beam reaches the tensile cracking stress of concrete

$$M_{cr} = Z(f'_{ct.f} - \sigma_{cs} + P / A_g) + Pe \geq 0$$

Z = section modulus uncracked section I_g / y

σ_{cs} = maximum shrinkage induced tensile stress on uncracked section at the extreme fibre at which cracking occurs taken as

$$\sigma_{cs} = E_s \varepsilon_{cs}^* (2.5 p_w - 0.8 p_{cw}) / (1 + 50 p_w)$$

Web reinforcement ratio for tensile reinforcement

$$p_w = (A_{st} + A_p) / (b_w d)$$

Web reinforcement ratio for compressive reinforcement

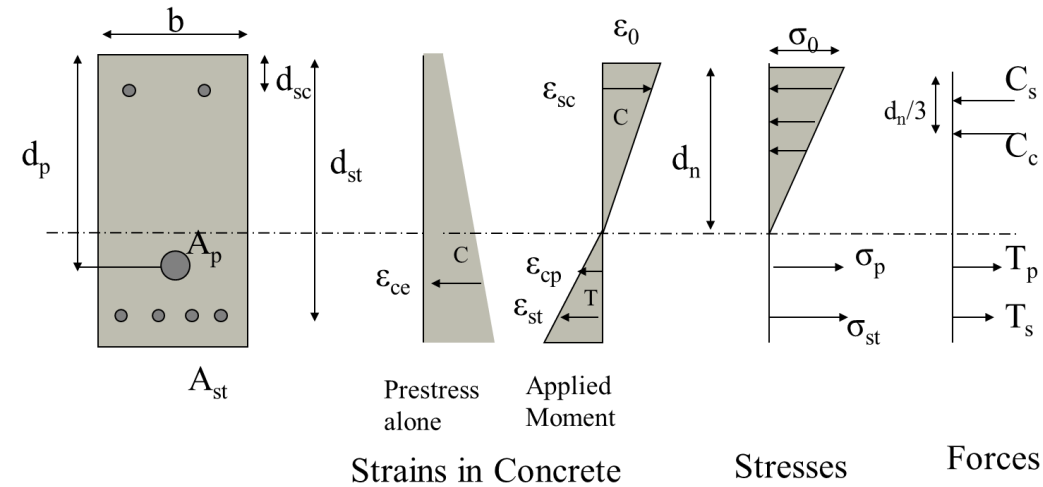
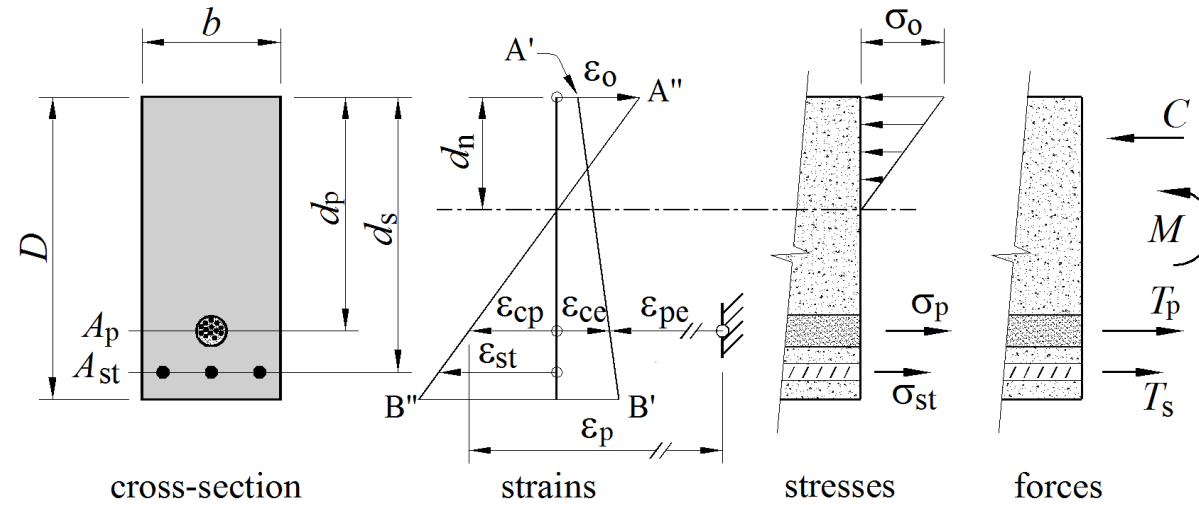
$$p_{cw} = A_{sc} / b_w d$$

ε_{cs}^* final design shrinkage strain of concrete

$f'_{cf.f}$ = flexural tensile strength of concrete

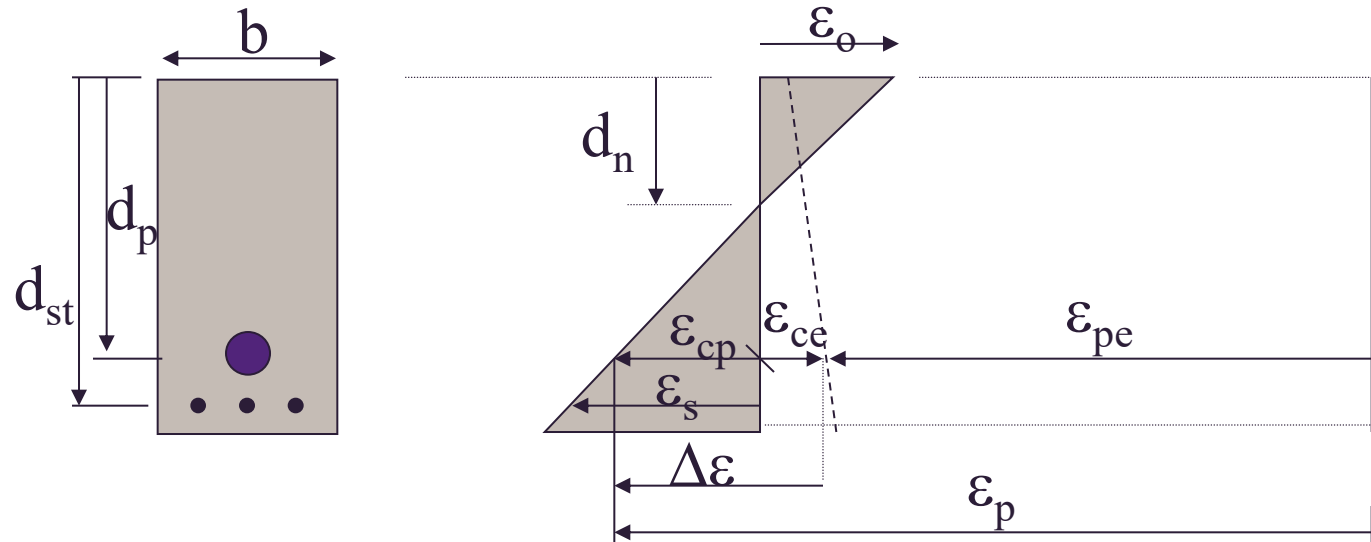
$$f'_{ct.f} = 0.6 \sqrt{f'_c}$$

Elastic Analysis of a rectangular cracked section



Determining strain in prestress steel

Strains in section with prestress



Due to prestress only, strain in concrete at tendon level is ϵ_{ce} (+ve); strain in the tendon is ϵ_{pe} (-ve)

Applied moment causes strain in concrete to change from +ve ϵ_{ce} to -ve ϵ_{cp} . The concrete at level of the tendon has therefore undergone a tensile strain increment $\Delta\epsilon = \epsilon_{cp} + \epsilon_{ce}$ (absolute values). Assuming perfect bond, the same change in strain must occur in tendon.

Total strain in prestress $\epsilon_p = \epsilon_{pe} + \epsilon_{ce} + \epsilon_{cp}$

Where $\epsilon_{cp} = \epsilon_o(d_p - d_n)/d_n$

Equations of equilibrium for the section

Force Equilibrium

$$C_c + C_s = T_p + T_s$$

Moment Equilibrium,

Taking moments about the top of the section, (anti-clockwise is +ve)

$$M = T_s d_{st} + T_p d_p - C_c d_n/3 - C_s d_{sc}$$

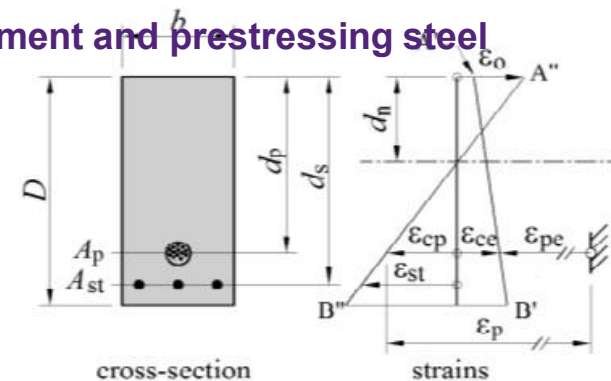
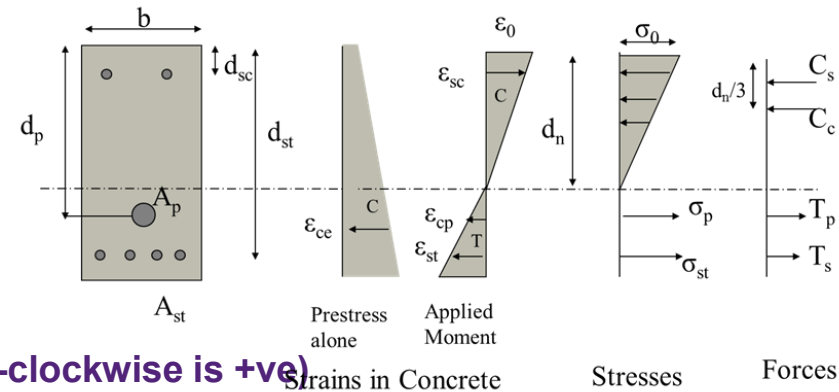
(line of action C_c is $d_c = d_n/3$)

Where 'M' is the applied external moment

Tensile strain in the concrete at the levels of the reinforcement and prestressing steel

$$\varepsilon_{st} = \varepsilon_{cst} = \varepsilon_0 (d_{st} - d_n) / d_n$$

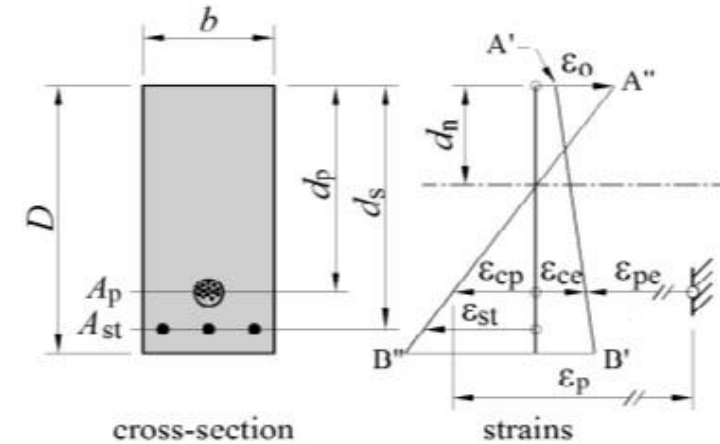
$$\varepsilon_{cp} = \varepsilon_0 (d_p - d_n) / d_n$$



Strain in prestressing steel

due to effective prestress

$$\varepsilon_{pe} = \sigma_p / E_p = P / (A_p E_p)$$



Compressive stress in concrete adjacent to the prestressing cable, due to prestress alone
(use $P = P_e$ = effective prestress force)

$$\varepsilon_{ce} = \sigma_{ce} / E_c \quad \text{where} \quad \sigma_{ce} = P / A_g + P e^2 / I_g$$

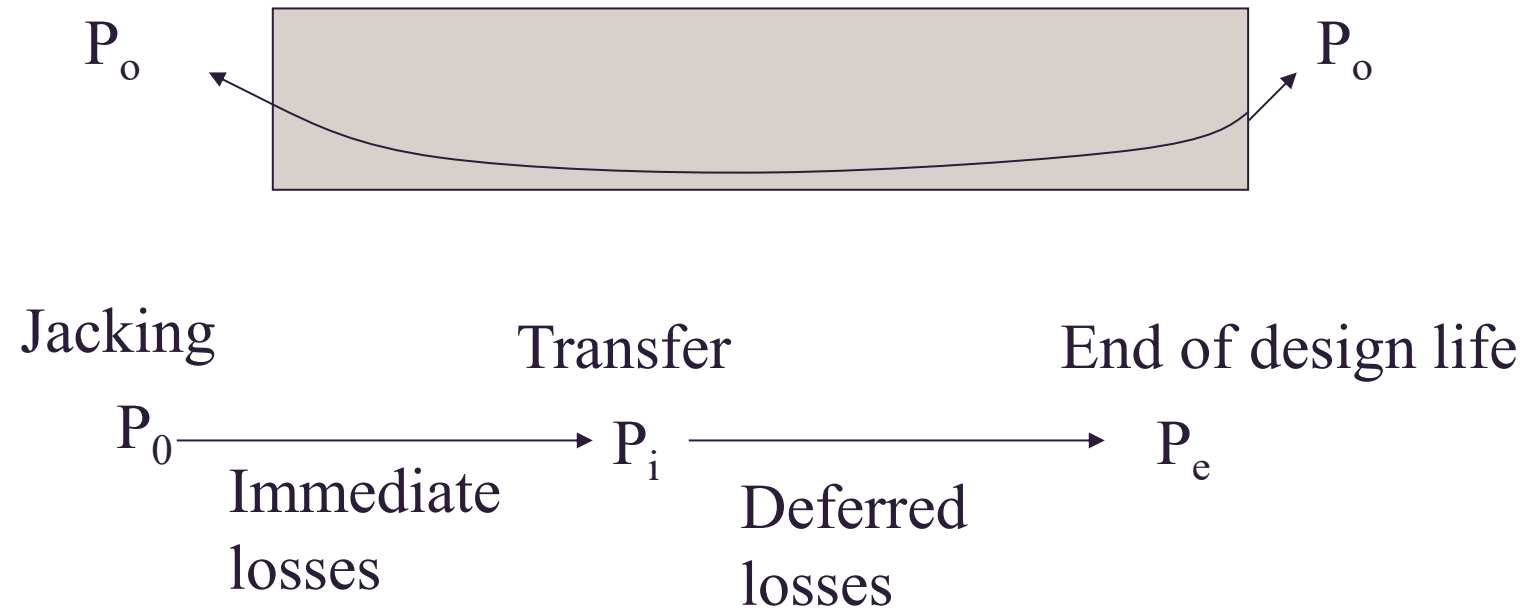
Tensile strain in concrete at level of prestress steel due to applied Moment

$$\varepsilon_{cp} = \varepsilon_o (d_p - d_n) / d_n$$

$$\text{Total strain in prestress steel} = \varepsilon_p = \varepsilon_{pe} + \varepsilon_{ce} + \varepsilon_o (d_p - d_n) / d_n$$

Losses of Prestress

Chapter 9 – Reference book Warner et al.



Contents

- Introduction to loss of prestressing
- Short-term loss (Immediate loss)
- Long-term loss (time-dependent loss)
- Examples and exercises

Introduction to Loss of Prestress

- The lack of success experienced in early attempts to prestress concrete was due to failure to appreciate the significance of the inevitable partial loss of prestress force.
- One of the design conditions is: At service – maximum bending moment + minimum prestress force. It is important, therefore, to obtain an estimate of the minimum prestress force throughout the structure.
- Prestressing force is not constant along the span of the tendons if the tendon is not straight. Therefore, in curved tendon profile, the prestressing force is depending on the position.
- There are a lot of factors affecting the prestressing force in strands after stressing.

Introduction to Loss of Prestress

- Some of these losses are immediate (short-term losses), which affects the prestressing force as soon as it is transferred to the concrete member.
- Some other losses will however increase with time (long-term losses)

<i>Short term</i>	<i>Long term</i>
Elastic shortening	Shrinkage
Friction	Creep
Anchorage draw-in	Prestressing steel relaxation

- Short-term losses depend on the stressing method (pre- or post-tensioning) and profile of strands/cables (straight or curved). Therefore, not all the above short-term losses will happen in prestressed concrete structures.

Introduction to Loss of Prestress

- Nevertheless, the long-term losses will basically occur in all types of prestressed concrete structures.
- High accuracy is rarely justified in determining the loss of prestress force and an accuracy of $\pm 10\%$ is sufficient for most purposes.
- Since loss of prestress varies in short and long terms, there are different effective prestressing force after short-term (immediately after transfer) and long-term losses (during service).
- Notation:
 P_o = applied prestressing force (at jacking) without any loss considered
 P_i = effective prestressing force immediately after all short-term losses have occurred at transfer stage
 P_e = effective prestressing force after long-term losses have occurred during service
- For the purpose of preliminary design, which should be verified in the detailed design stage, it is fairly reasonable to assume
 - 15% short-term loss and additional 10% long-term loss

Types of Losses

Short-term Losses (Immediate)

Elastic shortening (deformation) of the concrete, during stressing

Anchorage Slip, ie movement of the grips as the tendons are anchored

Duct Friction, ie friction along the member, between the tendon and duct, during stressing of the tendon

Deferred Losses

Stress relaxation in the prestressing steel

Shrinkage deformation in the concrete

Creep deformation in the concrete

In AS3600 other immediate losses are listed and may be considered in special circumstances. eg. deformation of construction joints in segmented members, temperature effects and deformation in forms for precast members

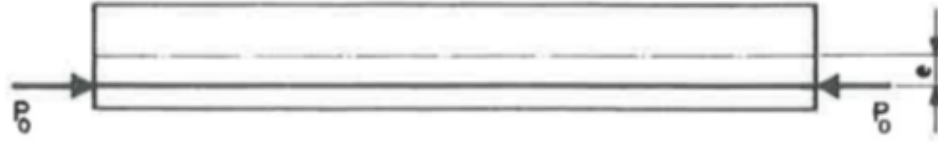
Short-Term Loss

Elastic Loss (Shortening)

- Elastic shortening of the concrete occurs when the prestressing force is transferred from the steel to concrete – because the concrete is subjected to compression.
- For ***pre-tensioned members***, in which the tendon is bonded to the concrete at the time of transfer, the change in steel strain is the same as the concrete compressive strain at the level of the steel centroid, and the loss may be calculated accordingly.
- The situation of pre-tensioned members will be discussed first and the equation will be applied to ***post-tensioned*** members differently.

Short-Term Loss

Elastic Shortening - Consider a pre-tensioned member with an eccentric prestress force P_o .



In a *pre-tensioned member* during transfer, the tensile strain in the prestressing steel decreases by an amount equal to the compressive strain induced by the concrete at the level of the steel, since the steel and concrete are bonded.

If σ_{ci} is the compressive stress at the steel level just after transfer, **the loss in prestress in the tendon, $\Delta\sigma_p$** due to elastic compression is:

$$\Delta\sigma_p = \sigma_{ci}E_p/E_c$$

Where the concrete stress σ_{ci} is determined from the initial prestressing force P_i

Elastic Loss

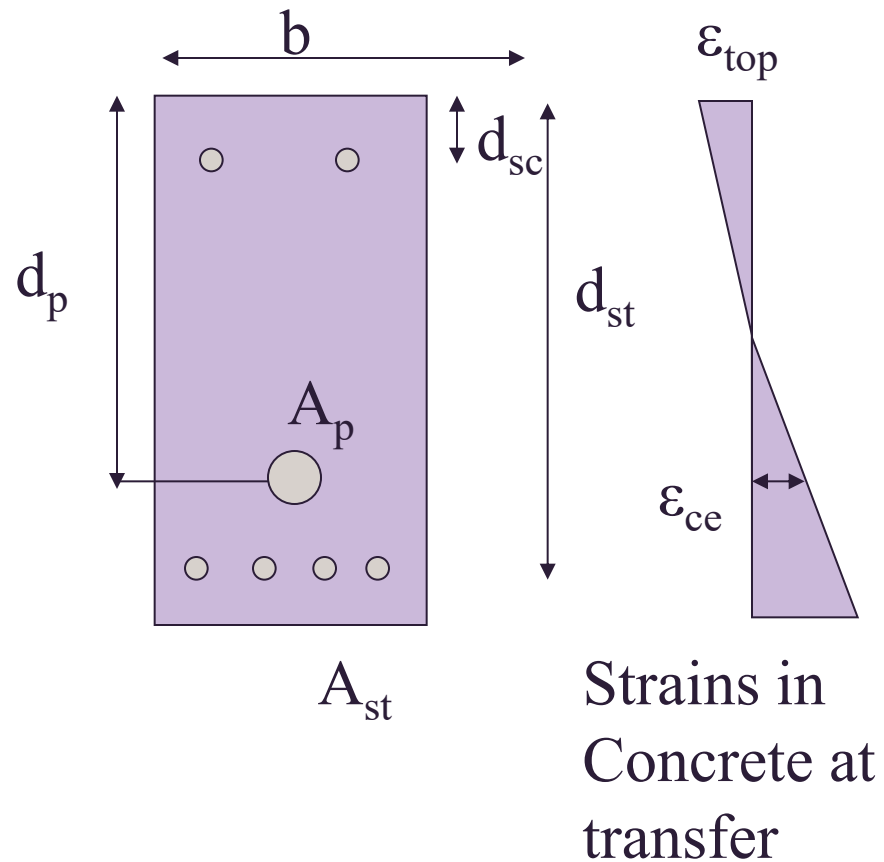
The concrete stress σ_{ci} is determined from the initial prestressing force P_i and setting y to the cable eccentricity, e .

$$\sigma_{ci} = \frac{P_i}{A_g} + \frac{P_i e^2}{I_g}$$

In order to achieve a predetermined prestressing force P_i just after transfer, the required force P_o in the cable prior to transfer is obtained combining the equation for σ_{ci} and the loss in prestress in the tendon $\Delta\sigma_p$

$$P_o = P_i \cdot \left[1 + \left(\frac{1}{A_g} + \frac{e^2}{I_g} \right) \cdot \frac{E_p}{E_c} \cdot A_p \right]$$

Elastic Shortening Loss – Single Cable



ϵ_{ce} = concrete compressive strain at the steel level just after transfer, (ie. due to prestress alone)

$$\epsilon_{ce} = \sigma / E_c$$

$$= 1/E_c [P_i/A_g + (P_i e^2)/I_g]$$

Reduction in Stress in the prestressing tendon

$$\Delta \sigma_p = \sigma_{ci} E_p / E_c = \epsilon_{ce} E_p$$

Reduction of force in the prestressing tendon = $\epsilon_{ce} E_p A_p$

Elastic loss- number of cables, sequential stressing

In a **post-tensioned member**, the cable is prestressed directly against the concrete, which compresses elastically as the cable force increases. *Provided there is only one cable, there is no need to make allowance for elastic loss.*

Where prestressing steel consists of a number of cables post-tensioned sequentially, condition more complex

Tensioning of one cable can cause losses in all the cables stressed earlier – this is common in post-tensioned beams

The first cable will undergo the largest loss of prestress

When the k^{th} cable at an eccentricity e_k is stressed to P_k , the elastic loss in a previously tensioned cable j , of area A_{pj} at an eccentricity e_j is

$$\Delta P_{kj} = P_k \cdot \left(\frac{1}{A_g} + \frac{e_k e_j}{I_g} \right) \cdot \frac{E_p}{E_c} \cdot A_{pj}$$

Note: tendons are bunched together and an approximate estimate of elastic loss can be obtained by using half the value of $\Delta\sigma_p$

Short Term Loss- Duct Friction

- For post-tensioned members, the tendons are usually anchored at one end and stretched with the jacks at the other end. As the steel slides through the duct, frictional resistance is developed, with the result that tension at the anchored end is less than the tension at the jack.
- Two types of friction
- Curvature friction, which is due to the non-linear profile of the tendons and is intentional. This is due to the induced normal force created by the curvature of the tendon profile. The normal force will then generate friction when the tendons are stressed, which is related by the coefficient of friction μ .

Short Term Loss- Duct Friction

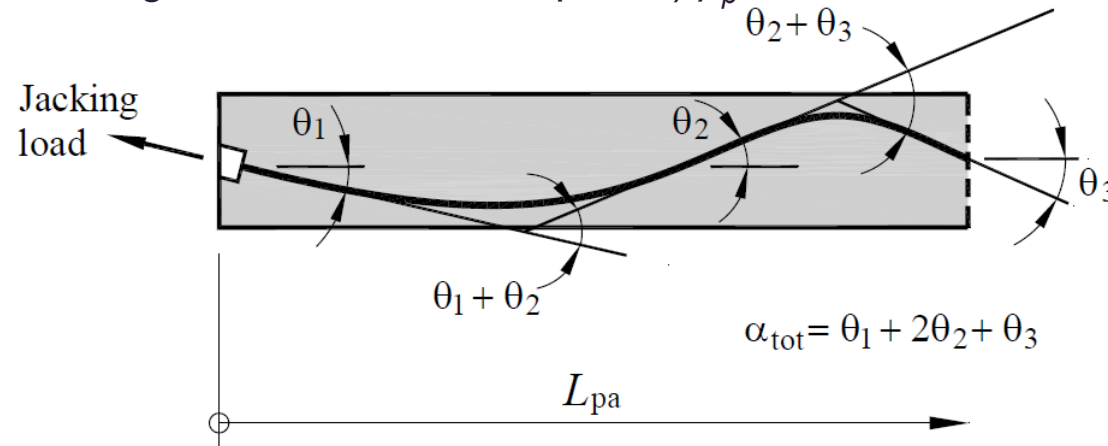
- As a result, straight tendon will have no loss with respect to friction.
- Wobbling effect - inevitable and unintentional deviation between the centrelines of the tendons and the ducts. This happens because in practice, the duct for the tendon cannot be perfectly straight.
- Some friction will exist between the tendon and its surrounding material even though the tendon is meant to be straight.
- Also, tendons are quite heavy. Under gravity, it is very hard to straighten the tendons even after stressing. This effect is normally referred to the wobbling effect (or wobbling friction).
- The total effect due to friction is the arithmetic sum of these components plus cable angular change

Short term - Duct Friction

Hence, in post tensioned members friction between the cable and surrounding duct produces a fall-off in prestressing force with distance from the jacking point, that is the prestress at midspan to be less than the jacking force

This loss is a function of

- -the length of the cable from jacking end to design section L_{pa} ,
- -total angular change over the length of the cable α_{tot}
- -Duct friction μ
- -Duct wobble (local irregularities of the cable profile) β_p



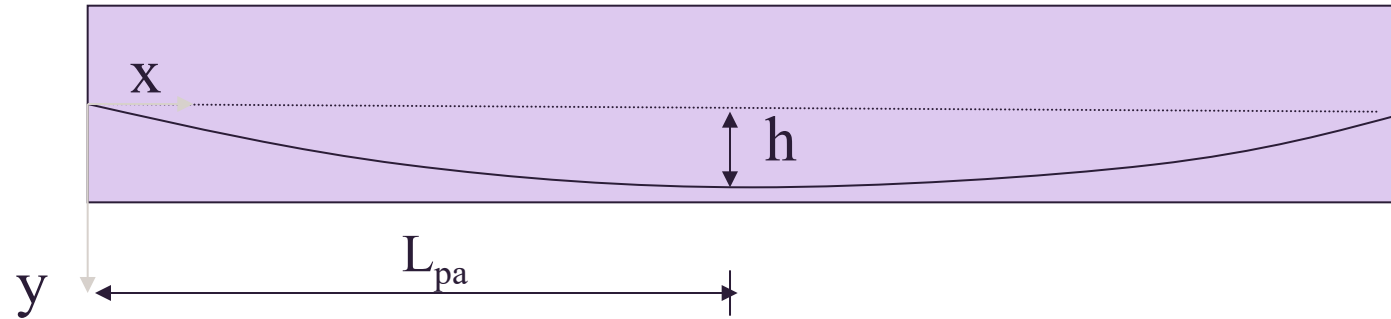
AS3600 method of calculation of duct friction loss

Cable stress σ_{pa} at a distance L_{pa} from the jacking end is calculated as

$$\sigma_{pa} = \sigma_{pj} e^{-\mu(\alpha_{tot} + \beta_p L_{pa})}$$

AS3600-2018 Cl.3.4.2.4 gives the factors μ and β_p for different types of ducts

Calculation of α_{tot}



α_{tot} = sum of all changes in angle in the cable over the distance L_{pa} .

For the example above, the equation of the cable profile is

$$y = 4h[x/L - (x/L)^2]$$

$$\text{Slope} = dy/dx = \theta = 4h/L[1 - 2(x/L)]$$

$$x = 0, \theta = 4h/L$$

Angular change between end anchorage and mid-span = $4h/L$

Short Term Loss- Anchorage Slip

Anchorage draw-in

- Clause 3.4.2.5 of AS 3600-2018
- For post-tensioning, when a tendon is tensioned to its full value, the jack is released and the prestress is transferred to the anchorage.
- The anchorage fixtures that are subjected to stresses at this transfer will tend to deform, thus allowing the tendon to slacken slightly.

Anchorage



Wedge

Short Term Loss- Anchorage Slip

- Friction wedges employed to hold the wires will slip a little distance before the wires can be firmly gripped. The amount of slippage depends on the type of wedge and the stress in the wires: **3 – 6 mm**.
- The slip of wedges can be reduced by ensuring that they are pushed forward as far as possible to grip the tendons before releasing the jack.
- Since the anchorage draw-in is a fixed amount depending on the type of anchorage used, the effect is much greater on a short prestressed concrete member than on a long one, because the effect of draw-in will affect a longer length of tendon.
- The anchorage draw-in effect in post-tensioning can be easily compensated by stretching longer the prestressing tendon by the predicted draw-in amount.

Short Term Loss- Anchorage Slip

- The full loss of prestress due to anchorage slip or wedge slip occurs at the jacking end of tendon. Due to friction in the duct, the slip loss diminishes with distance away from the jacking end and become zero at a point. Fig 9.2a)
- The loss due to anchorage slip is therefore often small at critical design sections near mid span of flexural member

Rate of strain loss: m_f

Strain loss AD

$$\Delta \varepsilon = 2m_f L_{ws} / E_p$$

Wedge-slip

$$\delta_{ws} = \Delta \varepsilon L_{ws} / 2 = m_f L_{ws}^2 / E_p$$

Wedge-slip length

$$L_{ws} = \sqrt{\frac{E_p \delta_{ws}}{m_f}}$$

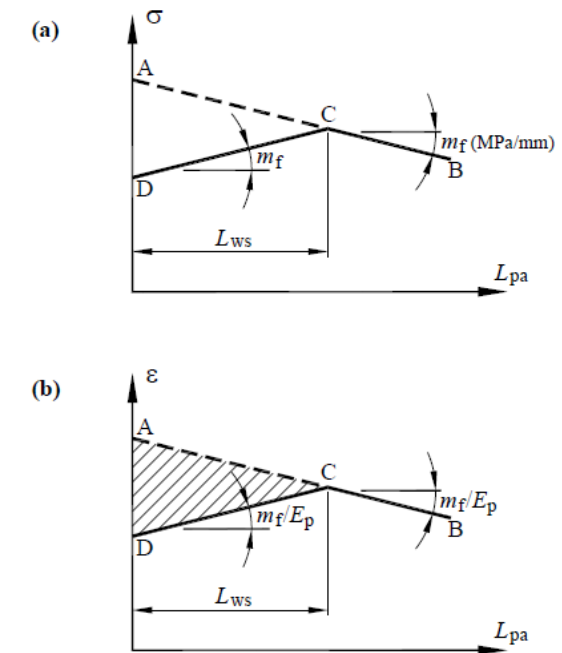


Figure 9.2 Anchorage slip loss

Deferred losses

Shrinkage deformation in the concrete

Creep deformation in the concrete

Stress relaxation in the prestressing steel

Loss of prestress due to shrinkage of concrete

The loss in prestress due to a design shrinkage strain ε_{cs} in the concrete is

$$\Delta\sigma_p = E_p \varepsilon_{cs}$$

and modified to allow for the effects of reinforcement as given in AS3600 - Cl3.4.3.2

The *free* shrinkage strain in the concrete at any time T after casting is estimated from shrinkage data provided in AS3600 Cl 3.1.7.2 –Design shrinkage strain

Actual field shrinkage will be less than the free shrinkage if any physical restraints acts on the concrete.

For example, if a prestressed member has reinforcement distributed throughout its cross-section so that its effect on shrinkage is mainly axial, the loss of prestress in the tendon is modified using a factor α calculated as $[1+15A_s/A_g]$ hence $\Delta\sigma_p = E_p \varepsilon_{cs} / \alpha$

Calculation of design shrinkage strain ϵ_{cs}

Shrinkage of concrete occurs due to evaporation of water from drying concrete

Design Shrinkage strain ϵ_{cs} is calculated as the sum of the chemical (autogenous) shrinkage strain (ϵ_{cse}) and the drying shrinkage strain (ϵ_{csd})

$$\epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd}$$

AS3600-2018 Cl 3.1.7.2

Autogenous shrinkage strain, ε_{cse}

Chemical (or autogenous) shrinkage results from various chemical reactions within the cement paste and includes hydration shrinkage, which is related to the degree of hydration of the binder in a sealed specimen.

$$\varepsilon_{cse} = \varepsilon_{cse}^* \times (1.0 - e^{-0.07t})$$

Where t is time in days after setting and ε_{cse}^* is the final autogenous shrinkage strain given by

$$\varepsilon_{cse}^* = (0.07f'_c - 0.5) \times 50 \times 10^{-6} \text{ for } f'_c \text{ less than or equal to } 50 \text{ MPa}$$

$$\varepsilon_{cse}^* = (0.08f'_c - 1.0) \times 50 \times 10^{-6} \text{ for } f'_c \text{ greater than } 50 \text{ MPa}$$

Drying shrinkage strain, ϵ_{csd}

Drying shrinkage is the reduction in volume caused principally by the loss of water during the drying process.

At any time t (in days) after the commencement of drying, the drying shrinkage strain shall be taken as

$$\epsilon_{csd} = k_1 k_4 \epsilon_{csd.b}$$

3.1.7.2 k_1 depends on the dimensions of the member calculated as $t_h = 2A/u_e$ and obtained from figure

Where A is the cross sectional area of the member

u_e is the length of the exposed perimeter

The basic drying shrinkage strain $\epsilon_{csd.b}$ is given by

$$\epsilon_{csd.b} = (0.9 - 0.005f'_c) \times \epsilon_{csd.b}^*$$

$\epsilon_{csd.b}^*$ depends upon quality of the local aggregates and taken as 800×10^{-6} or by testing

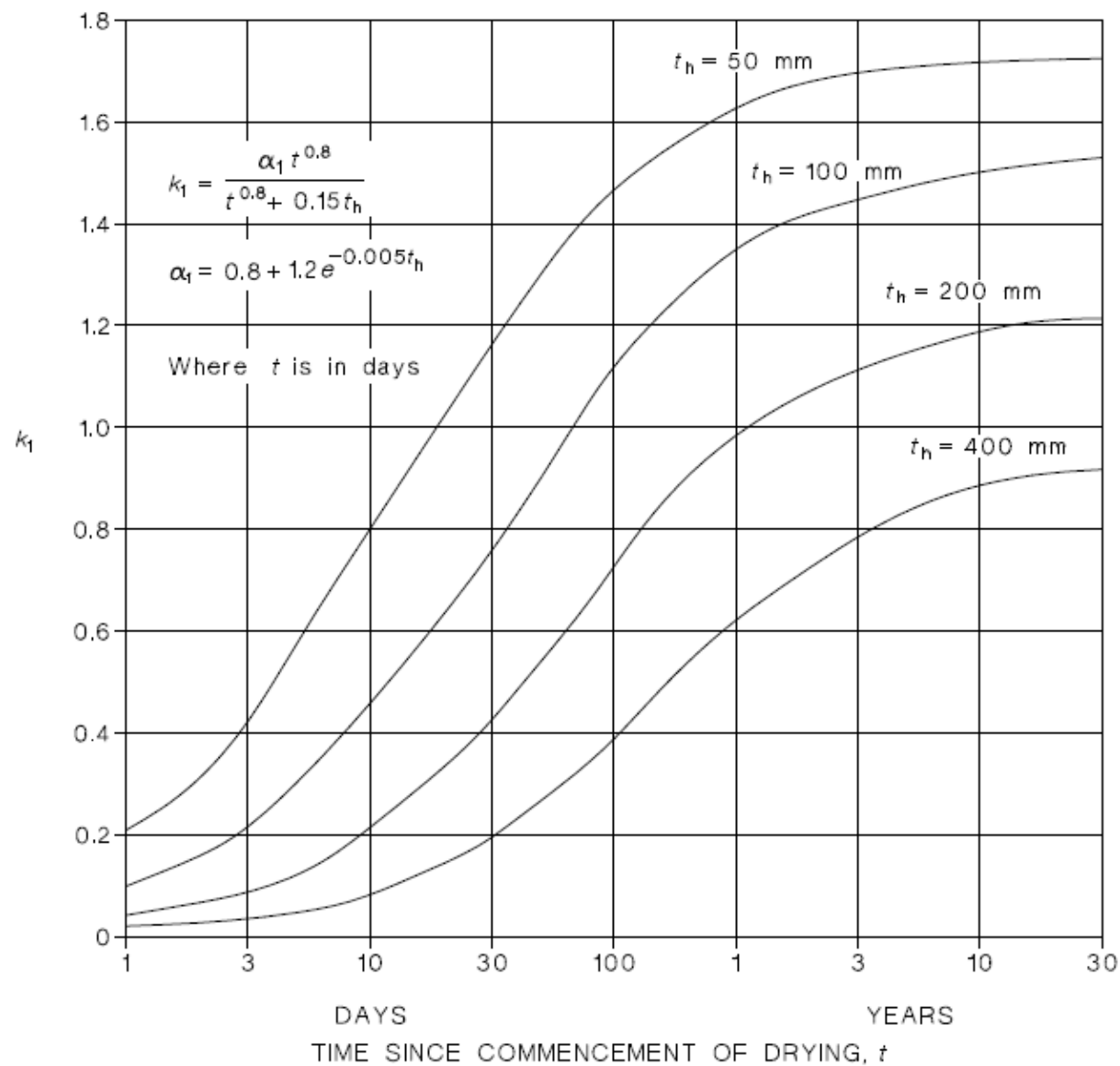


FIGURE 3.1.7.2 SHRINKAGE STRAIN COEFFICIENT (k_1) FOR VARIOUS VALUES OF t_h

Typical final design shrinkage strain after 30 years.

TABLE 3.1.7.2
TYPICAL FINAL DESIGN SHRINKAGE STRAINS AFTER 30 YEARS

f'_c (MPa)	Final design shrinkage strain $\epsilon_{cs}^s (\times 10^{-6})$															
	Arid env ⁺ [No Title] ^t				Interior environment				Temperate inland environment				Tropical, near-coastal and coastal environment			
	t_h (mm)				t_h (mm)				t_h (mm)				t_h (mm)			
	50	100	200	400	50	100	200	400	50	100	200	400	50	100	200	400
25	810	720	590	470	760	670	550	440	710	630	520	410	600	530	440	350
32	800	720	590	470	750	670	560	440	700	630	520	420	600	540	450	360
40	790	710	590	480	740	670	560	450	700	620	530	430	600	540	460	380
50	780	700	590	490	730	660	560	460	690	620	530	440	600	540	470	390
65	770	700	600	510	730	660	570	490	690	630	550	470	610	560	490	420
80	750	690	610	530	720	660	590	510	680	630	560	490	620	570	510	460
100	740	690	620	560	710	660	600	540	680	640	580	530	630	590	540	500

Loss of prestress due to creep of concrete

The loss in stress in the prestressing steel due to creep strain ε_{cc} in the concrete is calculated as in Cl3.4.3.3

$$\Delta\sigma_p = E_p \varepsilon_{cc}$$

creep strain of concrete is

$$\varepsilon_{cc} = 0.8\phi_{cc}(\sigma_{ci}/E_c) \quad \text{Cl3.4.3.3 of AS3600}$$

Where the concrete stress σ_{ci} is the sustained stress in the concrete at the level of the centroid of the tendons, calculated using the initial prestress force P_i prior to any time-dependent losses and the sustained portions of all service loads

ϕ_{cc} = design creep coefficient at time (t) determined in accordance with Cl 3 .1.8.3 and depends on

t_h and Relative humidity (RH) as in shrinkage

f'_c characteristic strength of concrete

$\phi_{cc.b}$ basic creep coefficient

$$\phi_{cc} = k_2 \times k_3 \times k_4 \times k_5 \times k_6 \phi_{cc.b}$$

Design creep coefficient ϕ_{cc}

ϕ_{cc} = design creep coefficient at time (t)

$$\phi_{cc} = k_2 \times k_3 \times k_4 \times k_5 \times k_6 \phi_{cc.b}$$

k_2 and k_3 obtained from fig 3.1.8.3

$k_3 = 2.7 / [1 + \log (\tau)]$ $\tau \geq 1$ day

$k_4 = 0.7$ for arid environment, 0.65 for interior environment, 0.6 for temperate, 0.5 for tropical

k_5 modification for high strength concrete, taken as 1.0 for $f'_c < 50\text{MPa}$

$k_6 = 1.0$

$\phi_{cc.b}$ basic creep coefficient given in table 3.1.8.2

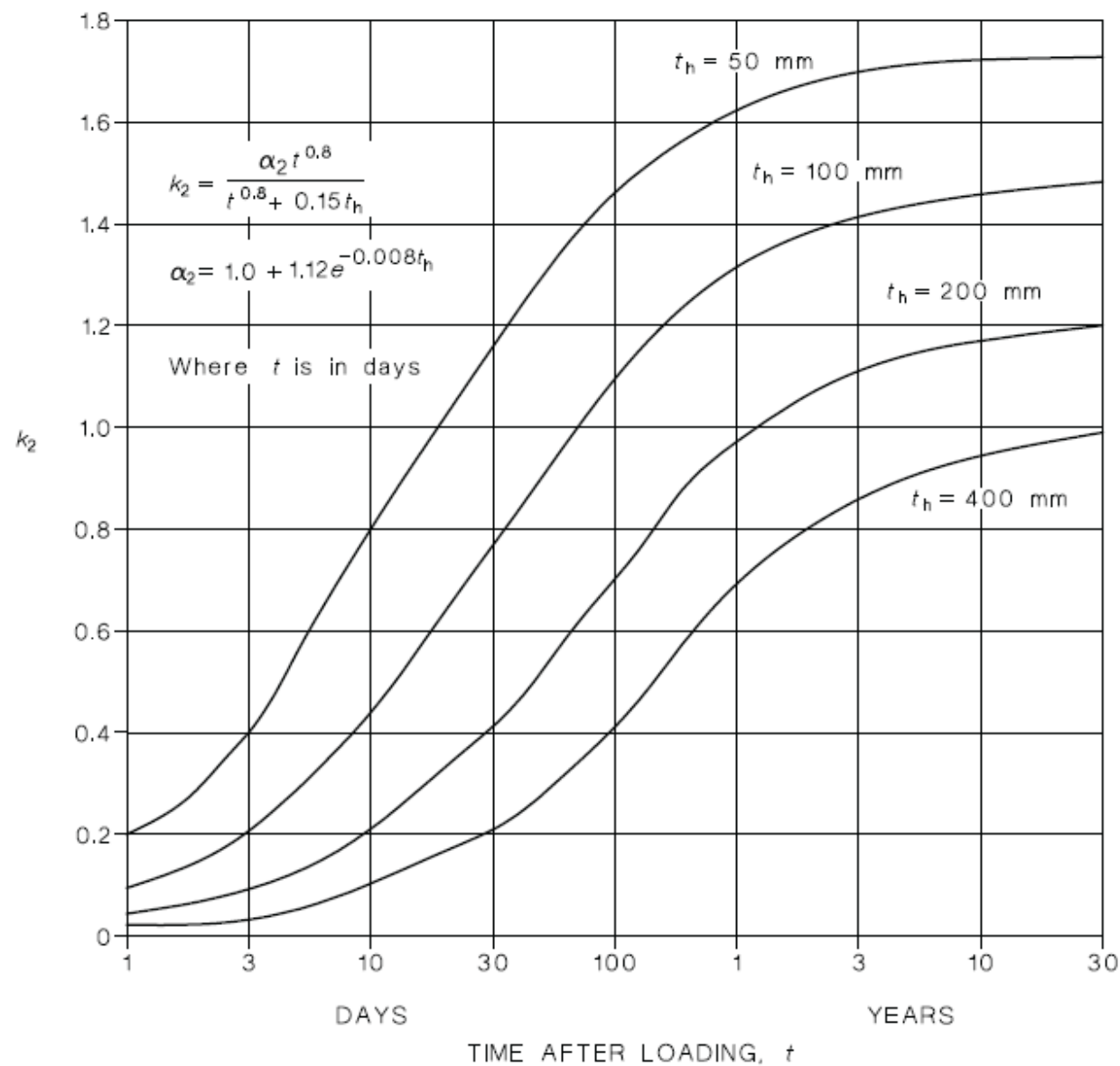


FIGURE 3.1.8.3 COEFFICIENT (k_2)

TABLE 3.1.8.2
BASIC CREEP COEFFICIENT

Characteristic strength (f'_c), MPa	20	25	32	40	50	65	80	100
Basic creep coefficient ($\phi_{cc,b}$)	5.2	4.2	3.4	2.8	2.4	2.0	1.7	1.5

--

Typical final creep co-efficient after 30 years.

TABLE 3.1.8.3
FINAL CREEP COEFFICIENTS (AFTER 30 YEARS)
FOR CONCRETE FIRST LOADED AT 28 DAYS

f'_c (MPa)	Final creep coefficient (ϕ_{cc}^+)											
	Arid environment			Interior environment			Temperate inland environment			Tropical, near-coastal and coastal environment		
	t_h (mm)			t_h (mm)			t_h (mm)			t_h (mm)		
	100	200	400	100	200	400	100	200	400	100	200	400
25	4.82	3.90	3.27	4.48	3.62	3.03	4.13	3.34	2.80	3.44	2.78	2.33
32	3.90	3.15	2.64	3.62	2.93	2.46	3.34	2.70	2.27	2.79	2.25	1.90
40	3.21	2.60	2.18	2.98	2.41	2.02	2.75	2.23	1.87	2.30	1.86	1.56
50	2.75	2.23	1.89	2.56	2.07	1.73	2.36	1.91	1.60	1.97	1.59	1.33
65	2.07	1.75	1.53	1.95	1.66	1.46	1.84	1.59	1.38	1.61	1.38	1.23
80	1.56	1.40	1.29	1.50	1.36	1.25	1.45	1.32	1.22	1.33	1.23	1.14
100	1.15	1.14	1.11	1.15	1.14	1.11	1.15	1.14	1.11	1.15	1.14	1.11

Loss of prestress due to tendon relaxation

This is the creep in strands, which would cause a loss of stress

is determined by modifying the percentage loss of stress due to the design relaxation R to take into account effects of shrinkage and creep.

The percentage loss of stress in the tendon in the member is taken as (Cl3.4.3.4)

$$R(1 - (\text{the loss of stress due to creep and shrinkage} / \sigma_{pi}))$$

σ_{pi} = stress in tendon immediately after transfer

R = is design relaxation of tendon

Design relaxation of tendon R

Design relaxation of a tendon R is determined from Cl3.3.4.3

$$R \text{ (as \%)} = k_7 k_8 k_9 R_b$$

Basic relaxation is the assumed relaxation 1000 hours after stressing for a prestress level of $0.8f_{pb}$. Cl 3.3.4.2 for wire taken as percentage given in table on next slide

k_7 , k_8 k_9 are coefficient that account for, respectively, duration of the prestressing force, the level of stress in tendon and the temperature

$$k_7 = \log[5.4(j)^{1/6}]$$

j = time after prestressing in days

k_8 depends on the stress in the tendon as a proportion of f_p

k_9 generally taken as 1

Design relaxation of tendon R

TABLE 2.2 Relaxation data for prestressing steels (AS/NZS 4672.1)

Type of tendon	Initial force as percentage of f_{pb}	# Basic Relaxation R_b (max %)	
		Relax 1 (normal-relaxation)	Relax 2 (low-relaxation)
Stress relieved wire	80	10.0	3.0 #
	70	6.5	2.0
	60	3.5	1.0
7-wire strand	80	12.0	3.5 #
	70	8.0	2.5
High strength alloy bar	70	—	4.0 #
	60	—	1.5

Note: # Values of the Basic Relaxation, R_b , are used in Equation 2.1 below.

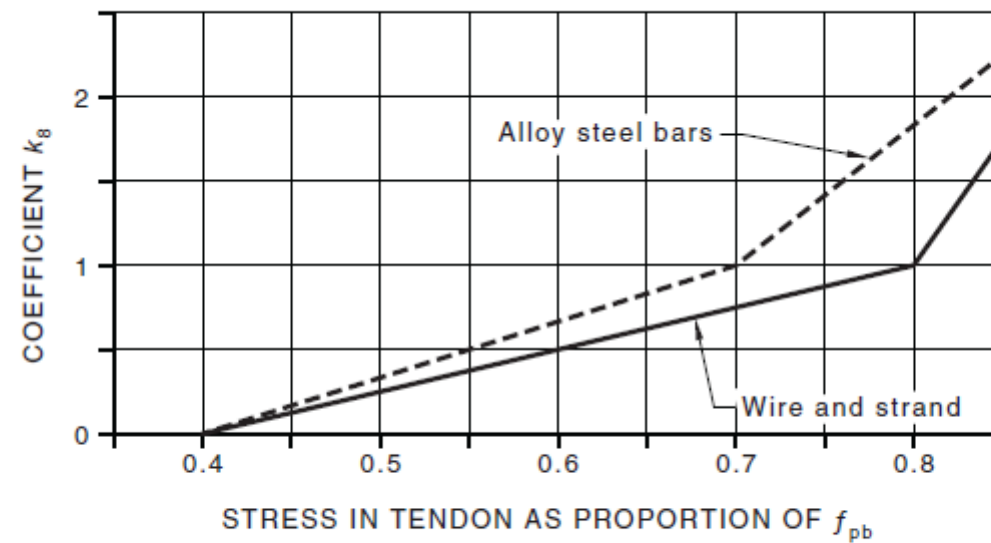
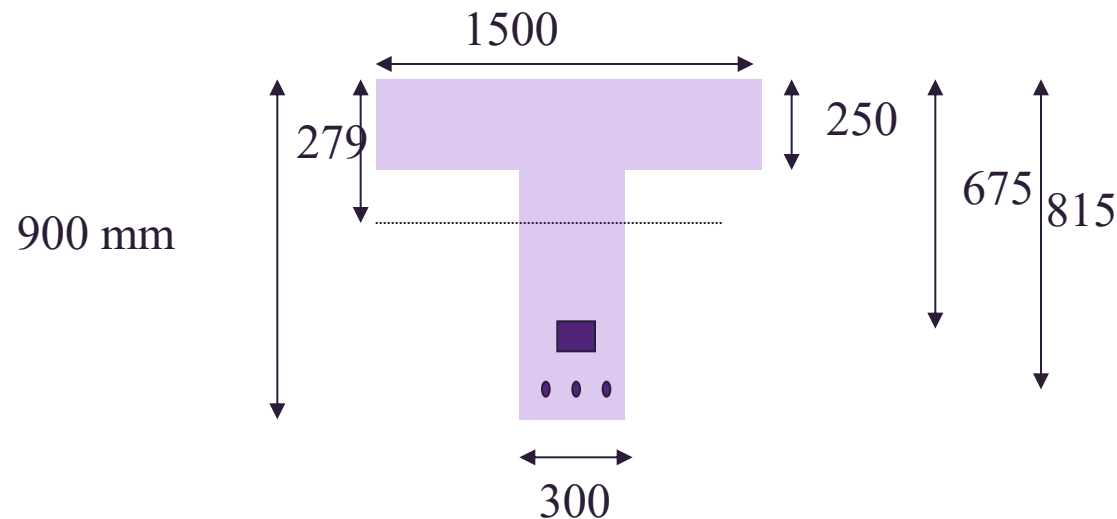


FIGURE 3.3.4.3 COEFFICIENT k_8

Worked example- Losses of prestress

Losses are to be determined for a simply supported post-tensioned beam which spans 25 m and has the cross-section shown in the figure. The beam carries no permanent load other than its weight.

Prestressing steel is contained in two cables each consisting of 18 super grade three wire low relaxation strands of 12.7 mm diameter. For each cable, the area is 1802 mm². The cables have parabolic profiles with eccentricities of zero at each end and 396 mm at mid-span. The concrete strength is $f'_c = 45$ MPa. Stressing takes place when the concrete is at age 14 days with a strength of 32 MPa. The beam is near coastal area.



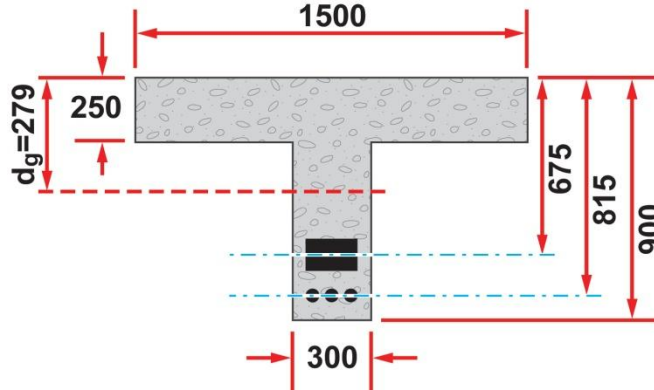
$$\sigma_{pi} = 1100 \text{ MPa}$$

$$A_p = 3604 \text{ mm}^2$$

$$A_s = 8000 \text{ mm}^2$$

$$A_g = 570,000 \text{ mm}^2$$

$$I_g = 3.48 \times 10^{10} \text{ mm}^4$$



$$\sigma_{pi} = 1100 \text{ MPa}$$

$$A_p = 3604 \text{ mm}^2 \text{ (two cables)}$$

$$A_s = 8000 \text{ mm}^2$$

$$A_g = 570,000 \text{ mm}^2$$

$$I_g = 3.48 \times 10^{10} \text{ mm}^4$$

$$f'_c = 45 \text{ MPa}$$

Beam exposed near – costal area

Immediate losses

Elastic Loss

The concrete compressive stress at the steel level at mid-span, due to the stressing of one cable is:

Eccentricity cable @ mid span = 396mm

Stressing takes place @ 14 days $f'_{cp} = 32 \text{ MPa}$ hence $f'_{cmi} = 35 \text{ MPa}$, $E_{cp} = 30100 \text{ MPa}$ – AS3600 Tb 3.1.2

$$\begin{aligned} \sigma_{ci} &= \frac{P_i}{A_g} + \frac{P_i e^2}{I_g} \quad \text{where } P_i = \sigma_{pi} \times A_p \text{ (one cable)} = 1100 \times 1802 \\ &= \frac{1100 \times 1802}{570,000} + \frac{1100 \times 1802 \times 396^2}{3.48 \times 10^{10}} \\ &= 3.478 + 8.93 \\ &= 12.4 \text{ MPa} \end{aligned}$$

The loss of prestress in the first cable when the second cable is stressed is:

$$\Delta \sigma_p = \sigma_{ci} \frac{E_p}{E_c} = 12.4 \times \frac{190,000}{30100} = 78 \text{ MPa}$$

The first cable could thus be stressed to $1100 + 80 = 1180 \text{ MPa}$ to allow for concrete elastic compression.

On stressing the second cable, the total area of prestressing steel is 3604 mm^2 and the average elastic loss is $78/2 = 39 \text{ MPa}$

Friction loss – Tendon

It is assumed that jacking is carried out at one end only (from Cl 3.4.2.4 AS3600 – 2018).

$$\beta_p = 0.016 \text{ rad} / m$$

(for sheathing with tendons of internal diameter $\leq 50 \text{ mm}$)

$\mu = 0.20$ (zinc-coated flat metal ducts)

At mid-span

$$\alpha_{\text{tot}} = \frac{4h}{L} = \frac{(4 \times (675 - 279))}{25,000} = 0.063 \text{ rad}$$

$$\sigma_{pa} = \sigma_{pj} e^{-u(\alpha_{\text{tot}} + \beta_p L_{pa})} = \sigma_{pj} e^{-0.2(0.063 + 0.016 \times 12.5)} = 0.95 \sigma_{pj}$$

At dead end

$$\alpha_{\text{tot}} = 2 \times 0.063 = 0.126 \text{ rad}$$

$$\sigma_{pa} = \sigma_{pj} e^{-0.2(0.126 + 0.016 \times 25)} = 0.91 \sigma_{pj}$$

Anchorage Slip

This is obtained from manufacturer's information. For example, for VSL multi-strand systems a draw-in loss of 6 mm is given.

Rate of strain loss: m_f

loss in first half of beam = $0.05 \times 1100 = 55 \text{ MPa}$

$$m_f = 55 / 12.5 = 4.4 \text{ MPa/m} \text{ (0.0044 MPa/mm)}$$

Wedge-slip length

$$L_{ws} = \sqrt{\frac{E_p \delta_{ws}}{m_f}} = \sqrt{\frac{190000 \times 6}{0.0044}} = 16.1 \times 10^3 \text{ mm}$$

Stress loss at the anchorage due to wedge slip

$$\Delta \sigma = 2 m_f L_{ws} = 2 \times 4.4 \times 16.1 = 142 \text{ MPa}$$

This reduces along the tendon length because of friction and at the mid-span ($L_{pa} = 12.5 \text{ metres}$) section the loss is approx.

$$142 \times (16.1 - 12.5) / 16.1 = 32 \text{ MPa}$$

which is about = 3% of the initial prestress

Deferred losses Shrinkage loss

Design shrinkage strain ϵ_{cs}

$$\epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd} \quad \text{Cl 3.1.7.2(1) AS3600 - 2009}$$

Autogenous shrinkage strain ϵ_{cse}

$$\epsilon_{cse} = \epsilon_{cse}^* \times (1.0 - e^{-0.07t})$$

Cl 3.1.7.2(2)

where t is the time (in days) after setting.

Take $t = 30$ years expected life-span of bridge = 10,950 days

Final autogenous shrinkage strain ϵ_{cse}^*

$$\begin{aligned} \epsilon_{cse}^* &= (0.07f'_c - 1.0) \times 50 \times 10^{-6} \quad \text{Cl 3.1.7.2(3)} \\ &= (0.07 \times 45 - 1.0) \times 50 \times 10^{-6} = 133 \times 10^{-6} \end{aligned}$$

$$\epsilon_{cse} = 133 \times 10^{-6} \times 1.0 - e^{(-0.1 \times 10950)} = 133 \times 10^{-6}$$

Drying shrinkage strain ϵ_{csd}

$$\epsilon_{csd} = k_1 k_4 \epsilon_{csd.b} \quad \text{Cl 3.1.7.2(4)}$$

$k_1 = 1.14$ figure 3.1.7.2 AS3600 - 2009

$$\left[\begin{array}{l} \text{for } t = 10950 \text{ days} \\ th = \frac{2 \times A_g}{U_e} = \frac{2 \times 570000}{2 \times (1500 + 900)} = 238 \text{ mm} \end{array} \right]$$

$k_4 = 0.5$ near-coastal area

$$\epsilon_{csd.b}^* = 800 \times 10^{-6} \quad \epsilon_{csd.b} = (0.09 - 0.005f'_c) \cdot \epsilon_{csd.b}^* = 540 \times 10^{-6}$$

$$\therefore \epsilon_{csd} = 1.14 \times 0.5 \times 540 \times 10^{-6} = 308 \times 10^{-6}$$

The design shrinkage strain is:

$$\epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd} = (133 + 308) \times 10^{-6} = 441 \times 10^{-6}$$

Loss of prestress due to shrinkage:

$$\Delta\sigma_p = \epsilon_{cs} \frac{E_p}{\alpha} \quad \text{Cl.3.4.3.2 AS3600 - 2018}$$

α = modification to allow for reinforcement effect

$$\alpha = 1 + 15 \frac{A_s}{A_g} = 1 + \frac{15 \times 8000}{570000} = 1.21$$

$$\Delta\sigma_p = 441 \times 10^{-6} \times \frac{190000}{1.21} = 69.2 \text{ MPa} \approx 69 \text{ MPa}$$

Creep loss

creep strain of concrete is

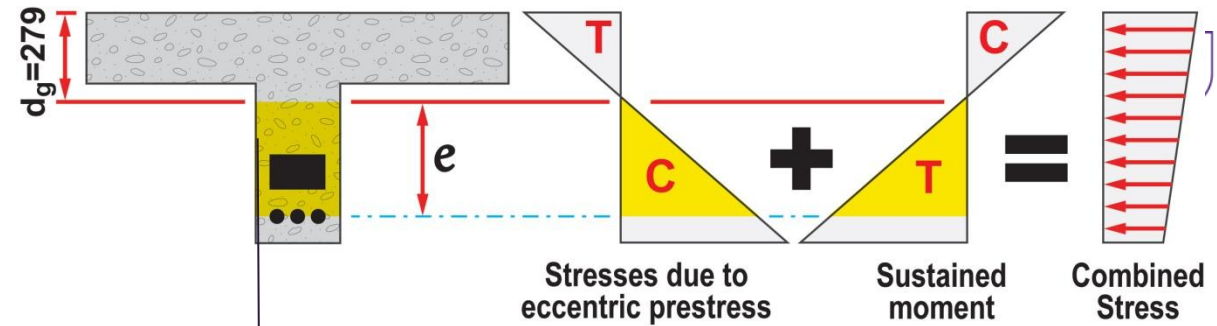
$$\varepsilon_{cc} = 0.8\phi_{cc} \left(\frac{\sigma_{ci}}{E_c} \right) \quad \text{Cl 3.4.3.3} \quad \text{AS3600}$$

σ_{ci} = sustained stress in concrete at level of tendons

Assuming that only sustained load is the self-weight = 13.7 kN/m the mid-span bending moment due to sustained load:

$$M_{sw} = \frac{13.7 \times 25^2}{8} = 1070 \text{ kNm}$$

The stress in concrete at the level of prestressing steel due to prestress and sustained service load.



$$[\text{at prestress level}] \sigma = \left(\frac{P_i}{A_g} + \frac{P_i e^2}{I_g} \right)$$

$$P_i = 1802 \times 1100 \times 2 = 3965 \text{ kN}$$

The resultant stress at the level of prestressing

$$\sigma_{ci} = \left[\frac{3965 \times 10^3}{570000} + \frac{3965 \times 10^3 \times 396^2}{34.8 \times 10^9} \right] - \frac{1070 \times 10^6 \times 396}{34.8 \times 10^9}$$

$$= 24.8 - 12.2 = 12.6 \text{ MPa}$$

Design creep co-efficient:

$$\phi_{cc} = \phi_{cc.b} k_2 k_3 k_4 k_5 k_6 \quad \text{Cl 3.1.8.3}$$

$$\phi_{cc.b} = 2.6 \quad (\text{Tb 3.1.8.2})$$

$$t = 30 \text{ years} \quad t_h = 238 \text{ mm}$$

$$k_2 = 1.143 \quad (\text{Figure 3.1.8.3})$$

$$k_3 = 2.7 / [1 + \log(t)] = 1.103$$

where t = 28 days when sustained load first applied

$$k_4 = 0.5$$

$$k_5 = 1.0 \quad f'_c \leq 50 \text{ MPa}$$

$$k_6 = 1.0$$

$$\therefore \phi_{cc} = 2.6 \times 1.143 \times 1.103 \times 0.5 \times 1.0 \times 1.0 = 1.64$$

The creep strain:

$$\varepsilon_{cc} = 0.8\phi_{cc} \left(\frac{\sigma_{ci}}{E_c} \right) = 0.8 \times 1.64 \left(\frac{12.6}{33800} \right) = 0.00049$$

Hence loss of prestress due to creep of concrete:

$$\Delta\sigma_p = \varepsilon_{cc} E_p = 0.00049 \times 190000 = 93.1 \text{ MPa}$$

$\Rightarrow$ sum of the losses due to creep and shrinkage:

$$\Delta\sigma_p = 69 + 93.1 = 162 \text{ MPa}$$

$$\text{per cent loss} = \frac{162}{1100} \times 100 = 14.7\%$$

Tendon stress relaxation

For low relaxation strand take

$$R_b = 3\%$$

Design relaxation of a tendon

R is

Cl 3.3.4.3 AS3600 – 2009

$$R = k_7 k_8 k_9 R_b$$

$$k_7 = \log \left(5.4(j)^{1/6} \right) = \log \left(5.4(10950)^{1/6} \right) = 1.4$$

$$k_8 = 0.5 \quad \left(\text{for } \frac{\sigma_{pi}}{f_p} = \frac{1100}{1870} = 0.6 \right)$$

$$f_{pb} = \text{characteristic minimum breaking strength (Cl 3.3.1)} = 1870 \text{ MPa}$$

$$k_9 = 1.0$$

Hence:

$$R = 1.4 \times 0.5 \times 1.0 \times 3 = 2.1\%$$

$$\begin{aligned} \text{loss of prestress} &= R(1 - \text{loss of stress due to creep and shrinkage}) / \sigma_{pi} \\ &= 2.1(1 - ((69 + 93)/1100)) = 1.79\% \end{aligned}$$

Summary of percentage losses @ mid-span section:

Elastic: $39/1100 \approx 4\%$

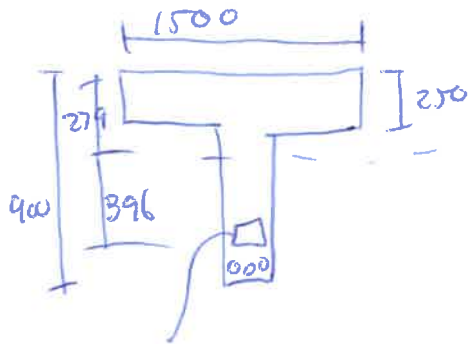
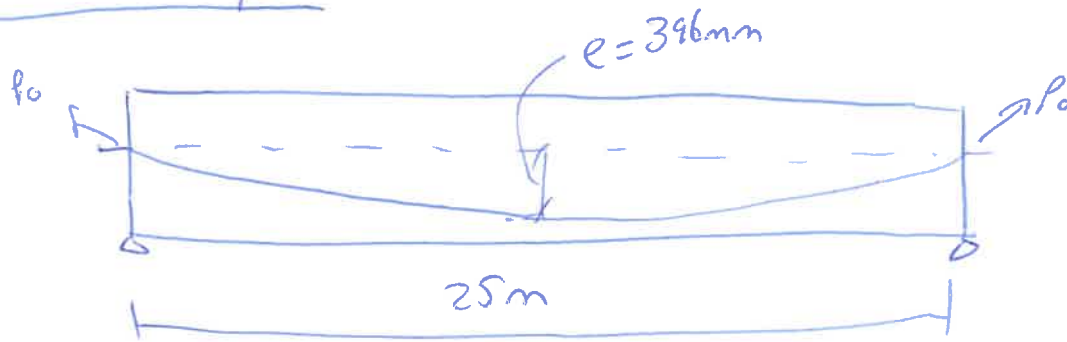
Friction: 5%

Anchorage Slip = 3%

Creep and shrinkage: 14.7%

Relaxation: 1.79%

Lecture example



two cables.

$$\sigma_{pi} = 1100 \text{ MPa}$$

$$A_p \text{ per cable} = 3604 \text{ mm}^2 / 2 = 1802 \text{ mm}^2$$

$$A_s = 8000 \text{ mm}^2$$

$$A_g = 570,000 \text{ mm}^2$$

$$I_g = 3.48 \times 10^{10} \text{ mm}^4$$

$$E_c = 30,000 \text{ MPa} \quad f'_{cp} = 32 \text{ MPa}$$

Short-term losses

→ Elastic loss.

the concrete compressive stress at the steel level at mid-span, due to the stressing of one cable is:

$$\sigma_{ci} = \frac{P_i}{A_g} + \frac{P_i \cdot e^2}{I_g}$$

where $P_i = \sigma_{pi} \times A_p$ (one cable)

$$P_i = (1100 \times 1802) \text{ kN}$$

$$\sigma_{ci} = \frac{(1100 \times 1802)}{570,000} + \frac{(1100 \times 1802) \times 396^2}{3.48 \times 10^{10}}$$

$$= 3.48 + 8.93$$

$$= 12.4 \text{ MPa}$$

the loss of prestress in the first cable when second cable is stressed is;

$$\Delta \sigma_p = \sigma_{ci} \frac{E_p}{E_{cp}}$$

$$= 12.4 \times \frac{190,000}{30,100} = 78 \text{ MPa}$$

the first cable could thus be stressed to $1100 + 78 = 1178 \text{ MPa}$ to allow for concrete elastic compression.

On stressing the second cable, the total area of prestressing steel is 3804 mm^2 and the average elastic loss is $78/2 = 39 \text{ MPa}$.

⇒ Friction loss - Tendon

It is assumed that jacking is carried out at one end only (from Cl 3.4.2.4 - AS 3600)

$\beta_p = 0.016 \text{ rad/m}$ (for sheathing with tendons of internal diameter $\leq 50 \text{ mm}$)

$\mu = 0.2$ (zinc-coated flat metal ducts)

@ mid-span

$$\alpha_{\text{tot}} = \frac{4h}{L} = \frac{4 \times (675 - 279)}{25,000} = 0.063 \text{ rad.}$$

$$\sigma_{pa} = \sigma_{pj} e^{-\mu(\alpha_{\text{tot}} + \beta_p L_p)}$$

$$= \sigma_{ps} e^{-0.2(0.063 + 0.016 \times 12.5)} = 0.95 \sigma_{pj}$$

@ dead-end

$$\Delta_{tot} = 2 \times 0.063 = 0.126 \text{ rad.}$$

$$\sigma_{pa} = \sigma_{pj} e^{-0.2(0.126 + 0.016 \times 25)} = 0.91 \sigma_{pj}$$

$\Rightarrow$ Anchorage slip

this is obtained from manufacturer's information.
For example, for VSL multi-strand systems a draw-in loss of 6mm is given. However at the critical mid-span section, this loss will be ~~neglected~~ small.
(Anchorage slip calculated on pg 8)

Deferred Losses

$\Rightarrow$ shrinkage loss $f_{lc} = 45 \text{ MPa.}$

Design shrinkage strain, ϵ_{cs}

$$- \epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd} \quad \text{--- cl 3.17.2(1) AS3600}$$

Autogenous shrinkage strain, ϵ_{cse} --- cl 3.17.2(2) AS3600

$$- \epsilon_{cse} = \epsilon_{cse}^* \times (1.0 - e^{-0.07t})$$

t = time in days after setting, Take $t = 30 \text{ years}$
expected life-span of bridge = 10,950 days.

ϵ_{cse}^* = final autogenous shrinkage strain.

$$= (0.07 f_{lc} - 0.5) \times 50 \times 10^{-6}$$

$$= (0.07 \times 45 - 0.5) \times 50 \times 10^{-6} = 133 \times 10^{-6}$$

$$\epsilon_{cse} = 133 \times 10^{-6} \times (1.0 - e^{-767})$$

$$= \underline{133 \times 10^{-6}} \quad (\text{Auto}).$$

drying shrinkage strain, ϵ_{csd}

$$- \epsilon_{csd} = K_1 K_4 \epsilon_{csd,b} \quad - \text{cl 3.17.2 (4)}$$

$$K_1 = 1.14 \quad -$$

fig 3.17.2

for $t = 30$ years

$$t_h = \frac{2 \times A_g}{U_e} = \frac{2 \times 570,000}{2 \times (1500 + 900)} = 238 \text{ mm}$$

$K_4 = 0.5$ (near-coastal area)

$$\epsilon_{csd,b} = 800 \times 10^{-6} \quad \epsilon_{csd,b} = (0.9 - 0.005 f_{ck}) \times \epsilon_{cs,b} = 540 \times 10^{-6}$$

$$\epsilon_{csd} = 1.14 \times 0.5 \times 540 \times 10^{-6} = 308 \times 10^{-6} \quad (\text{dry})$$

$\Rightarrow$ the design shrinkage strain is:

$$\epsilon_{cs} = \epsilon_{cse} + \epsilon_{csd} \quad (133 + 308) = 441 \times 10^{-6}$$

- Loss of prestress due to shrinkage

$$\Delta \sigma_p = \epsilon_{cs} \frac{E_p}{\alpha} \quad - \text{cl 3.4-3.2}$$

α = modification to allow for reinforcement effect

$$\alpha = 1 + 15 \frac{A_s}{A_g} = 1 + \frac{15 \times 8000}{570,000} = 1.21$$

$$\Delta \sigma_p = 441 \times 10^{-6} \times \frac{190,000}{1.21} = 69.2 \text{ MPa}$$

⇒ Creep loss

creep strain of concrete

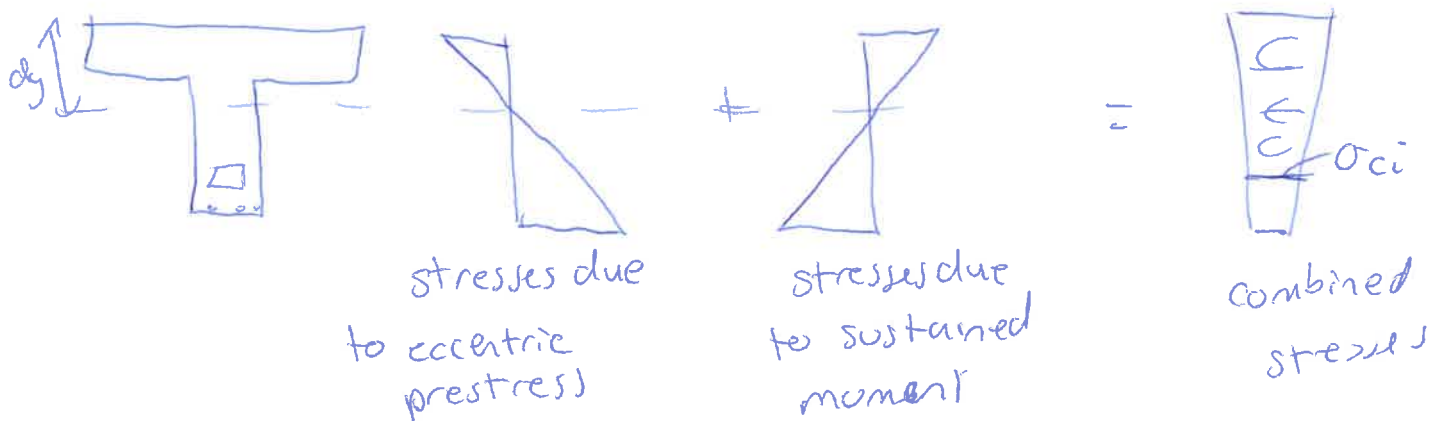
$$\epsilon_{cc} = 0.8 f_{cc} \left(\frac{\sigma_{ci}}{E_c} \right) \quad - \text{cl. 3.4.3.3}$$

σ_{ci} = Sustained stress in concrete at level of tendons.

Assuming that only sustained load is self-weight = 13.7 kN/m, the mid-span bending moment due to sustained load:

$$M_{sw} = \frac{w l^2}{8} = \frac{13.7 \times 25^2}{8} = 1070 \text{ kNm}$$

the stress in concrete at level of prestressing steel due to prestress and sustained service load.



$$\sigma_{ci} = \frac{P_i}{A_g} + \frac{P_i e^2}{I_g} - \frac{M_{sw} \cdot e}{I_g} \quad \text{where } P_i = 1802 \times 1100 \times 2 = 3965 \text{ kN}$$

$$= \left[\frac{3965 \times 10^3}{570,000} + \frac{3965 \times 10^3 \times 396^2}{34.8 \times 10^9} \right] - \frac{1070 \times 10^6 \times 396}{34.8 \times 10^9}$$

$$= 24.8 - 12.2 = 12.6 \text{ MPa.}$$

Design creep co-efficient

$$\phi_{cc} = k_2 k_3 k_4 k_5 k_6 \phi_{cc-b} \quad - \text{CI 3.1-8-3}$$

$$\phi_{cc-b} = 2.6 \quad \text{CTb 3.1-8-2}$$

$$k_2 = 1.143 \quad \left(t = 30 \text{ years}, t_h = 238 \right) \\ \text{fig. 3.1-8-3}$$

$$k_3 = 2.7 \left[1 + \log(t_0) \right] = 0.103$$

$t_0 = 28$ days when sustained load first applied

$$\phi_{cc} = 2.6 \times 1.143 \times 0.103 \times 0.5 \times 1 \times 1 = 1.64$$

$$(k_4 = 0.5 \quad k_5 = 1 \quad k_6 = 1)$$

- the creep strain:

$$\epsilon_{cc} = 0.8 \phi_{cc} \left(\frac{\sigma_{ci}}{E_c} \right) \quad E_c @ 45 \text{ MPa} = 33,800 \text{ MPa} \\ \text{CTb 3.1-2}$$

$$\epsilon_{cc} = 0.8 \times 1.64 \left(\frac{12.6}{33,800} \right)$$

$$= 0.00049$$

Hence loss of prestress due to creep of concrete

$$\Delta \sigma_p = \epsilon_{cc} E_p = 0.00049 \times 190,000 = 93.1 \text{ MPa}$$

$\Rightarrow$ sum of the losses due to creep + shrinkage

$$\Delta \sigma_p = 69 + 93.1 = 162 \text{ MPa}$$

$$\text{per cent loss} = \frac{162}{1100} \times 100 = 14.7\%$$

Tendon stress relaxation.

for low relaxation strand take.

$$R_b = 3\%$$

Design relaxation of tendon

$$R = K_1 K_2 K_3 R_b \quad \text{cl 3.3.4.3 of AS3600}$$

$$K_1 = \log(5.4(j)^{1/6}) = \log 5.4(10950)^{1/6} = 1.4$$

$$K_2 = 0.5 \quad \left(\text{for } \frac{\sigma_{pi}}{F_p} = \frac{1100}{1870} = 0.6 \right)$$

$$K_3 = 1.0$$

hence

$$R = 1.4 \times 0.5 \times 1.0 \times 3 = 2.1\%$$

$$\text{loss of prestress} = R \left(1 - \frac{(\text{loss of stress due to shrinkage \& creep})}{\sigma_{pi}} \right)$$

$$= 2.1 \left(1 - \left(\frac{(\cancel{69} + 93)}{1100} \right) \right)$$

$$= 1.79\%$$

Summary of percentage losses @ mid-span section:

$$\text{Elastic} = 39/1100 = 4\%$$

$$\text{Friction} = 5\%$$

$$\text{anchorage} = 3\%$$

$$\text{creep \& shrinkage} = 14.7\%$$

$$\text{Relaxation} = 1.79\%$$

Anchorage slip

- loss in first half of beam = $0.05 \times 1100 = 55 \text{ MPa}$
- rate of strain loss, m_f
 $m_f = 55/12.5 = 4.4 \text{ MPa/m}$ (0.0044 MPa/m)

- Wedge-slip length

$$L_{ws} = \sqrt{\frac{E_p \cdot S_{ws}}{m_f}} = \sqrt{\frac{190000 \times 6}{0.0044}} = 16.1 \times 10^3 \text{ mm}$$

- stress loss at the anchorage due to wedge slip.

$$\Delta \sigma = 2 m_f L_{ws} = 2 \times 4.4 \times 16.1 = 142 \text{ MPa}$$

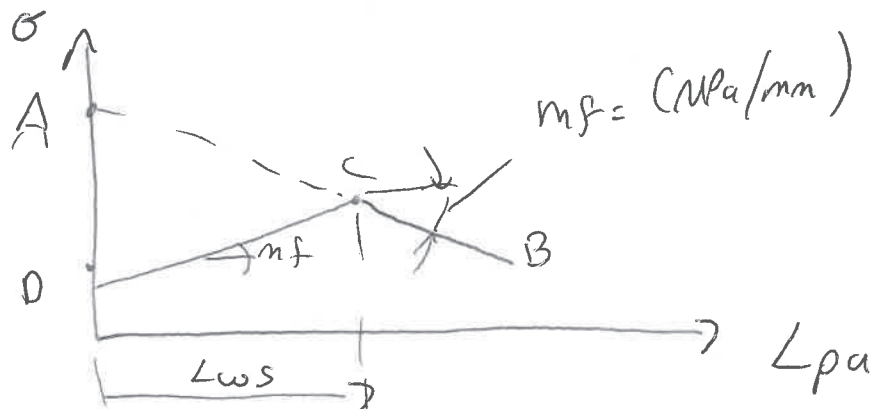
- this reduces along the tendon length because of friction and at the mid-span ($L_{pa} = 12.5 \text{ metres}$) section loss is approximately

$$142 \times (16.1 - 12.5) / 16.1 = 32 \text{ MPa}$$

which is about ~~1~~ 3% of the initial prestress

anchorage slip

- amount of slip depends on device $\approx 6\text{mm}$ draw-in.
- full loss of prestress due to anchorage slip occurs @ jacking end.
- Due to friction in the duct slip loss $\downarrow$ and at some point is zero

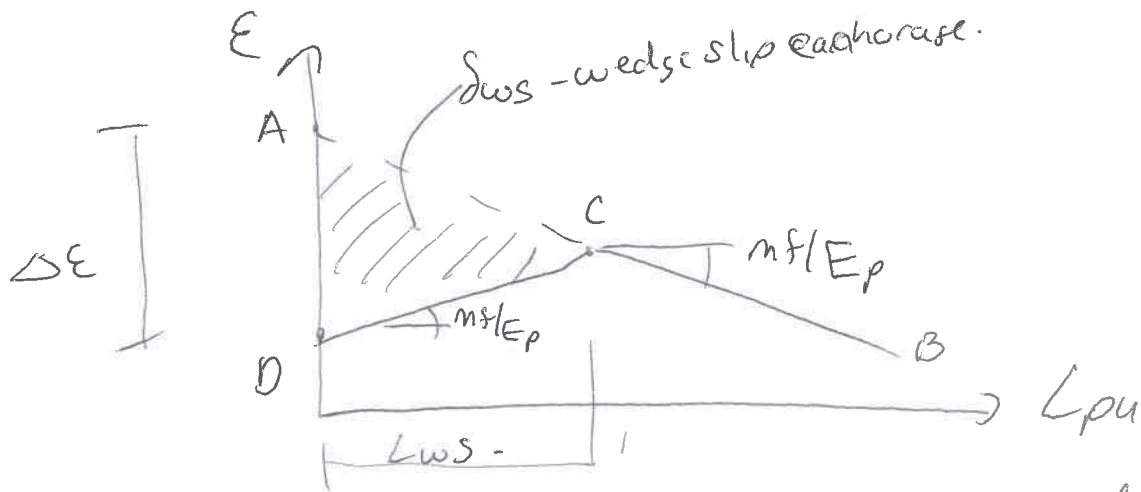


- AB represents prestress σ_{pi} in tendon; Maximum at jacking point A - reduces with distance due to friction loss, has linear slope

$$mf = (\text{MPa/mm})$$

- OC - ^{when} prestress force transferred from jack to the anchorage, the loss due to wedge draw-in is DC.
- two lines AB, AC have same slope.
- the length over which the cable force is affected by wedge slip is labelled L_{ws} .

- re-drawn now in terms of steel strain so that slope of lines AB & DC is now $m f / E_p$.
- the area ACD, must be equal to the wedge slip at the anchorage, δ_{ws} .



Hence L_{ws} can be expressed in terms of δ_{ws} .

$$\text{Strain loss } AD = \Delta E = 2mfL_{ws}/E_p$$

$$\text{Wedge slip : } \delta_{ws} = \frac{\Delta E L_{ws}}{2} = \frac{mf L_{ws}^2}{E_p}$$

$$\text{Wedge-slip length} = L_{ws} = \sqrt{\frac{E_p \delta_{ws}}{mf}}$$

$$\text{Prestress loss @ anchorage} = 2mf \cdot L_{ws}$$